SE-2317 ENTRY BLACKOUT STATE ESTIMATION

Gao Jiang

DOC NO:	TR-2026-001
DATE:	2026-05-09
REV:	A
STATUS:	DRAFT

ABSTRACT

This report presents the design and validation baseline of the SE-2317 entry blackout estimator. The implementation is a tightly-coupled Square-Root Error-State Kalman Filter with a fixed-lag history buffer for delayed and out-of-order measurements, replay-safe raw GNSS updates, configurable χ^2 gating, covariance inflation, optional innovation-based adaptive estimation, and pre-blackout accelerometer-bias calibration. The report summarizes the state model, observability considerations, delay handling method, numerical formulation, and Monte Carlo evidence carried by the repository. The March 11, 2026 benchmark numbers remain useful as a historical baseline, but they predate the current acceptance alignment and summary schema in the repository.

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DOCUMENT CONTROL

Program	SE-2317 — Entry Blackout Estimation
Doc Type	Technical Report
Revision	B (Draft)
Owner	Gao Jiang

Rev	Date	Author	Change Summary
A	2026-03-11	Gao Jiang	Initial report synchronized to the current estimator implementation and validation flow.
B	2026-05-09	Gao Jiang	Narrative, requirements wording, and validation summary aligned with the repository documentation set.
C	2026-06-13	Gao Jiang	Recovery-window reacquisition behavior, tight-coupling phase logic, and current quick-regression status synchronized to the repository state.

EXECUTIVE SUMMARY

Objective. Estimate 6-DOF entry state through a GNSS blackout window and meet the post-blackout (2 s) accuracy and consistency targets in `tasks.md`.
Deliverable. A documented estimator implementation with traceable mathematics, stable numerics, and reproducible Monte Carlo evidence.

Architecture summary. The repository implements a square-root error-state EKF on a 24-dimensional augmented state. The state is $(r, v, q, \omega, b_g, b_a, f_{\text{res}}, \tau_{\text{res}})$ with multiplicative quaternion error injection. Truth propagation and filter-side nominal linearization call the same aerodynamic and RCS composition helper. The state integrator continues to use de-biased IMU specific force directly.

Timing and robustness summary. Delayed and out-of-order measurements are handled by a fixed-lag history buffer with rewind–update–replay semantics covering GNSS PVT, raw GNSS, star tracker, and radar. The robustness layer combines χ^2 gating, Student-t-style covariance inflation, replay-safe IAE scaling, and pseudorange down-weighting for heavy-tailed residuals. Phase-dependent process-noise scaling and blackout-specific aerodynamic residual excitation are used to keep covariance realistic across mission phases. At blackout exit, the PVT path applies an immediate reacquisition update at the exit timestamp and uses the same velocity covariance as ordinary PVT tracking. In the tight-coupled path, the weakest-direction Doppler projection is confined to the two-second recovery interval after blackout exit, which preserves pre-blackout observability while still protecting the reacquisition segment against temporary geometry degradation. A pre-blackout accelerometer-bias calibration step is applied at blackout entry.

Validation summary. The historical release record dated 2026-03-11 reports a tight-coupled GNSS pass rate of **100.0%** (500/500) under 5% heavy-tail outliers and 20% dropout. The corresponding PVT baseline pass rate is **97.0%** (485/500). On the current code line, a release quick regression dated 2026-06-13 passes **10/10** in both modes. In that run, the recovery-window velocity three-sigma maximum is 0.415,7 m/s in PVT and 0.104,4 m/s in tight coupling. These numbers verify the present recovery-window fixes. A fresh long campaign is still required before promoting the current state to a new canonical validation record under the aligned acceptance logic.

1 PROBLEM SETTING AND REQUIREMENTS

The mission context is a 6-DOF rigid-body entry with strong aero nonlinearity and a GNSS blackout. Sensors are asynchronous with fixed delays (star tracker, radar altimeter) and stochastic dropouts and outliers. The primary acceptance window is the **2 seconds after blackout exit** ($t \in [t_{\text{exit}}, t_{\text{exit}} + 2]$, where $t_{\text{exit}} = t_0 + 160\,\text{s}$).

Key quantitative targets (summary). Within the post-blackout window: position $3\sigma \le 40\,\text{m}$, velocity $3\sigma \le 0.5\,\text{m/s}$, attitude $3\sigma \le 0.15^{\circ}$, bias bounds $\|b_g\|_{3\sigma} \le 0.005^{\circ}/\text{s}$ and $\|b_a\|_{3\sigma} \le 120\,\mu\text{g}$, and consistency via NEES/NIS 95% coverage ≥ 90% over Monte Carlo.

2 STATE, DYNAMICS, AND NOISE MODELING

2.1 State and error-state parameterization

We estimate the augmented state

$$x := \left[r,\ v,\ q,\ \omega,\ b_g,\ b_a,\ f_{\text{res}},\ \tau_{\text{res}}\right], \qquad r, v, \omega, b_g, b_a, f_{\text{res}}, \tau_{\text{res}} \in \mathbb{R}^3,\ q \in \mathbb{H},\ \|q\| = 1. \tag{1}$$

The EKF operates on an error state $\delta x \in \mathbb{R}^{24}$ with attitude error represented by a 3-vector $\delta\theta$ such that the corrected quaternion uses a multiplicative injection:

$$q^{+} = \exp(\tfrac{1}{2}\,\delta\theta) \otimes q^{-}, \qquad \exp(\tfrac{1}{2}\,\delta\theta) \approx \left[\cos(\tfrac{\|\delta\theta\|}{2}),\ \hat{\delta\theta}\sin(\tfrac{\|\delta\theta\|}{2})\right]. \tag{2}$$

This choice avoids quaternion normalization drift and yields a locally Euclidean error space for covariance.

2.2 Discrete-time propagation model

Let IMU measurements at time step k be $(\omega_{m,k}, a_{m,k})$. Bias-compensated signals are

$$\omega_k = \omega_{m,k} - b_{g,k}, \qquad a_k = a_{m,k} - b_{a,k}. \tag{3}$$

We propagate attitude by integrating ω_k over Δt and normalize:

$$q_{k+1} = \text{normalize}\Big(q_k \otimes \exp(\tfrac{1}{2}\,\omega_k \Delta t)\Big). \tag{4}$$

Define the body-to-inertial rotation $C(q) \in \text{SO}(3)$, and inertial gravity $g(r)$. The translational update is second-order in Δt:

$$v_{k+1} = v_k + \left(C(q_k)\,a_k + g(r_k) + a_{\text{learn}}(k)\right)\Delta t, \tag{5}$$

$$r_{k+1} = r_k + v_k \Delta t + \tfrac{1}{2}\left(C(q_k)\,a_k + g(r_k) + a_{\text{learn}}(k)\right)\Delta t^2. \tag{6}$$

Here a_{learn} is an optional bounded online correction (see §5). The nominal aerodynamic / RCS composition used by the filter Jacobian is generated by the same helper as truth propagation; the state update itself still uses de-biased IMU specific force so nominal terms are shared without double-counting the measured acceleration.

2.3 Process noise: IMU, bias RW, and aero residual OU

We model gyro/accel measurement noise as white with spectral densities σ_ω and σ_a, and biases as random walks:

$$b_{g,k+1} = b_{g,k} + w_{bg,k}, \quad b_{a,k+1} = b_{a,k} + w_{ba,k}, \quad w_{bg,k} \sim N(0, \sigma_{bg}^2 \Delta t\, I),\ w_{ba,k} \sim N(0, \sigma_{ba}^2 \Delta t\, I). \tag{7}$$

Aero residuals. To model dominant entry mismatch, we include residual force and torque states with Ornstein–Uhlenbeck (OU) discretization:

$$f_{\text{res},k+1} = \phi_f f_{\text{res},k} + \eta_{f,k}, \qquad \tau_{\text{res},k+1} = \phi_\tau \tau_{\text{res},k} + \eta_{\tau,k}, \quad \phi_f = e^{-\Delta t/\tau_f},\ \phi_\tau = e^{-\Delta t/\tau_\tau}, \tag{8}$$

with $\eta_{f,k} \sim N(0, \sigma_f^2(1-\phi_f^2)I)$ and $\eta_{\tau,k} \sim N(0, \sigma_\tau^2(1-\phi_\tau^2)I)$. This yields a stationary marginal variance $\sigma_f^2 I$ and $\sigma_\tau^2 I$.

Discrete translational acceleration noise. Assume a white specific-force disturbance affecting $\dot{v}$:

$$w_a \sim N(0, \sigma_{a,\text{eff}}^2 I). \tag{9}$$

The standard continuous-to-discrete integration gives the (RV) block:

$$Q_{\{r,v\}}(\Delta t) = \sigma_{a,\text{eff}}^2 \begin{bmatrix} \frac{\Delta t^3}{3} I & \frac{\Delta t^2}{2} I \\ \frac{\Delta t^2}{2} I & \Delta t\, I \end{bmatrix}. \tag{10}$$

We use $\sigma_{a,\text{eff}}^2$ (IMU noise plus model residual) and apply a phase-dependent scaling (pre/blackout/post) to enforce covariance realism where the model is weakest. We define the effective acceleration variance explicitly:

$$\sigma_{a,\text{eff}}^2 := \sigma_a^2 + \sigma_{\text{res}}^2. \tag{11}$$

3 OBSERVABILITY ANALYSIS AND BIAS MANAGEMENT

3.1 Local LTV observability (definition)

Let the linearized, time-varying error dynamics be

$$\delta x_{k+1} = F_k\, \delta x_k, \qquad y_k = H_k\, \delta x_k. \tag{12}$$

Define the state transition $\Phi_{k,0} = F_{k-1} \cdots F_0$. The system is observable on $k \in [0, N]$ if the stacked matrix $O_N := \begin{bmatrix} H_0 \\ H_1\Phi_{1,0} \\ \cdots \\ H_N\Phi_{N,0} \end{bmatrix}$ has full column rank. Equivalently, for weighted measurements with covariance R_k, the discrete observability Gramian

$$W_N := \sum_{k=0}^{N} \Phi_{k,0}^{\mathsf{T}} H_k^{\mathsf{T}} R_k^{-1} H_k \Phi_{k,0} \tag{13}$$

is symmetric positive definite. We use this definition to avoid over-claiming a closed-form condition in a strongly coupled entry model.

3.2 Unobservable subspace: Accelerometer bias and attitude coupling

The translational error-state dynamics contain both accelerometer bias and attitude error terms:

$$\delta v_{k+1} \approx \delta v_k - C_k\, \delta b_{a,k}\, \Delta t - \left[C_k a_k\right]_\times \delta\theta_k\, \Delta t + \cdots. \tag{14}$$

To show why bias is not generically identifiable without excitation, consider a simplified continuous-time, constant-excitation model with constant C and constant specific force f^n (in navigation frame), and ignore gyro-error dynamics:

$$\delta\dot{v} = -[f^n]_\times\, \delta\theta - C\, \delta b_a, \qquad \delta\dot{\theta} = 0, \qquad \delta\dot{b}_a = 0, \qquad y = \delta v. \tag{15}$$

Lemma. The above system is not observable; in particular, it has a non-trivial unobservable subspace.

Proof. Let $\delta\theta \neq 0$ be arbitrary and set $\delta b_a = -C^{-1}[f^n]_\times \delta\theta$ with initial $\delta v(0) = 0$. Then $\delta\dot{v}(t) = 0$, hence $y(t) = \delta v(t) \equiv 0$ for all t despite $\delta\theta \neq 0$ and $\delta b_a \neq 0$. Therefore the system cannot be observable. *QED*

Implication. Separating accelerometer bias from attitude error requires time variation (in C_k and/or in the direction of f) so that information accumulates in the LTV Gramian W_N. Entry dynamics provide such excitation, but only sensors that constrain δv (GNSS, tight-coupled Doppler) can turn that excitation into bias information.

3.3 Blackout implication and design response

During blackout, GNSS-derived constraints are absent; the bias sub-system loses direct observability, and any bias correction must come from (i) pre-blackout information, (ii) post-blackout reacquisition, and (iii) conservative covariance growth. The implementation therefore adopts two linked measures. A dedicated pre-blackout bias calibration reduces bias uncertainty before observability is lost. In parallel, blackout covariance realism is enforced through increased Q and stronger OU excitation in the aerodynamic residual states, so the filter does not carry an unrealistically tight prior into the recovery segment.

3.4 Pre-blackout accelerometer-bias calibration as MAP least squares

We treat a short pre-blackout interval as a dedicated calibration window. Let (r_k, v_k) be the estimator INS solution and $(\bar{r}_k, \bar{v}_k)$ the GNSS RV measurement at time t_k with covariance R_k. Linearize the RV residual with respect to a constant accelerometer-bias perturbation δb_a and initial RV offsets:

$$\begin{bmatrix} \bar{r}_k - r_k \\ \bar{v}_k - v_k \end{bmatrix} \approx \underbrace{\begin{bmatrix} I & (t_k - t_0)I & S_{r,k} \\ 0 & I & S_{v,k} \end{bmatrix}}_{A_k} \underbrace{\begin{bmatrix} \delta r_0 \\ \delta v_0 \\ \delta b_a \end{bmatrix}}_{\theta} + \varepsilon_k, \qquad \varepsilon_k \sim N(0, R_k), \tag{16}$$

where $(S_{r,k}, S_{v,k})$ are the accumulated RV sensitivities to δb_a along the propagated trajectory. We compute a MAP estimate with diagonal Gaussian priors $\theta \sim N(0, \Lambda^{-1})$:

$$\theta^* = \arg\min_\theta \sum_k \left\| R_k^{-1/2}(A_k\theta - b_k) \right\|^2 + \left\| \Lambda^{1/2}\theta \right\|^2, \qquad b_k := \begin{bmatrix} \bar{r}_k - r_k \\ \bar{v}_k - v_k \end{bmatrix}. \tag{17}$$

The solution satisfies the normal equations

$$\left(\sum_k A_k^\mathsf{T} R_k^{-1} A_k + \Lambda \right) \theta = \sum_k A_k^\mathsf{T} R_k^{-1} b_k. \tag{18}$$

Implementation notes: we (i) require a minimum observation count, (ii) enforce SPD via symmetrization and diagonal jitter, and (iii) clamp δb_a magnitude to a mission-tuned maximum step before applying it at blackout entry.

4 TIME ALIGNMENT AND DELAYED MEASUREMENTS

4.1 Model of delayed measurements

Let a sensor produce a measurement of the state at time t_z but deliver it at a later time $t_{\text{rx}} > t_z$. The correct Bayesian update is conditioned on $x(t_z)$:

$$p(x(t_z) \mid y_{0:\text{rx}}) \propto p(y(t_z) \mid x(t_z))\, p(x(t_z) \mid y_{0:t_z^-}). \tag{19}$$

The fundamental challenge is the temporal discrepancy between the filter state at update time t_{rx} and the valid state for the measurement $x(t_z)$.

4.2 Equivalence to stochastic cloning (linear-Gaussian)

For a linear Markov model, augment the state with a clone at t_z:

$$\tilde{x} := \begin{bmatrix} x(t_z) \\ x(t_{rx}) \end{bmatrix}, \qquad x(t_{rx}) = F\,x(t_z) + w, \;\; w \sim N(0, Q). \tag{20}$$

A delayed measurement depends only on the clone $x(t_z)$:

$$y(t_z) = H\,x(t_z) + v, \quad v \sim N(0, R). \tag{21}$$

Applying the Kalman update on $\tilde{x}$ yields the optimal posterior over $x(t_{rx})$ via the cross-covariance between $x(t_z)$ and $x(t_{rx})$. This corresponds to the usual stochastic-cloning fixed-lag smoothing construction.

4.3 Implementation: rewind–update–replay (fixed horizon)

Instead of explicitly augmenting the state, we maintain a fixed-lag buffer containing snapshots $(\hat{x}_k, L_k)$ at a coarse cadence such as 100 Hz, a complete event log for IMU and all measurement updates, and propagation records (Φ_k, Q_k) used by both RTS smoothing and replay. When a measurement arrives with timestamp $t_z < t_{now}$, the estimator restores the latest snapshot at $t_s \le t_z$, replays the logged events over $(t_s, t_z]$ to reconstruct $\hat{x}(t_z)$, applies the update at t_z, and then replays the remaining events over $(t_z, t_{now}]$ to return to the current state.

Correctness statement (linear case). For linear-Gaussian models with deterministic replay of the same event sequence, the rewind–update–replay procedure produces the same posterior as stochastic cloning. It reconstructs the correct conditional prior at t_z and then re-applies the same Bayes updates in time order. For EKF, this remains a first-order local linearization and is empirically stable under the tested delay and outlier regimes.

5 ROBUSTNESS AND CONSISTENCY CALIBRATION

5.1 Innovation statistics: NIS/NEES definitions

Let the innovation be $\tilde{y} = y - h(\hat{x}^-)$ with covariance $S = HPH^{\mathsf{T}} + R$. Define the (normalized) innovation squared:

$$\text{NIS} := \tilde{y}^{\mathsf{T}} S^{-1} \tilde{y}. \tag{22}$$

For a consistent filter and correct R, NIS $\sim \chi^2_v$ where v is the measurement DOF.

Let estimation error be $e = x - \hat{x}$ with covariance P. Define the normalized estimation error squared:

$$\text{NEES} := e^{\mathsf{T}} P^{-1} e. \tag{23}$$

For a consistent filter and correct (F, Q, R), NEES $\sim \chi^2_n$ for the chosen DOF n. We report both the mean and the coverage within the central 95% chi-square interval.

5.2 Chi-square gating (outlier rejection)

We accept a measurement update if

$$\text{NIS} \le \chi^2_\nu(1-\alpha), \tag{24}$$

where α is the gate significance (false-rejection probability). The gate limits the effect of sparse outliers on both the state and the covariance.

5.3 Heavy-tail mitigation for tight-coupled pseudorange

For raw GNSS pseudorange rows, we additionally apply a robust variance inflation: let the standardized residual for row i be $u_i = \tilde{y}_i/\sqrt{S_{ii}}$. For $|u_i| > k$, we inflate the measurement variance as

$$R_{ii} \leftarrow R_{ii}\left(\min\left(\frac{|u_i|}{k}, c\right)\right)^2, \tag{25}$$

with constants $k = 4$ and cap $c = 100$. The resulting weighting is equivalent to an M-estimator with a Huber-like influence function: large residuals receive reduced weight without forcing hard rejection.

5.4 Runtime robust tuning

The repository also exposes a runtime robustness surface used by the sweep harness: ‘`gate_p`‘ adjusts the effective χ^2 gate level, ‘`student_nu`‘ enables Student-t-style covariance inflation, ‘`r_scale_cap`‘ bounds covariance inflation, and ‘`iae_gamma`‘ enables replay-safe innovation-based adaptive estimation. Together these parameters define the tuning surface used by the sweep harness. The adaptive state is stored inside the fixed-lag snapshots so rewind/replay remains deterministic.

5.5 Consistency enforcement: phase-dependent Q and covariance inflation

Two mechanisms address the non-stationary model errors inherent to entry dynamics. The first is phase-dependent process-noise scaling, together with blackout-specific OU excitation for the aerodynamic residual states. The second is innovation-triggered covariance inflation after repeated GNSS gate failures, which acts as a guarded reset of the RV covariance. Mathematically, both increase P in regimes where NEES indicates under-dispersion, pushing $\mathbb{E}$[NEES] back toward the DOF.

6 NUMERICAL STABILITY: SQUARE-ROOT FORMULATION

We maintain the covariance in Cholesky form $P = LL^\mathsf{T}$ with L lower-triangular. Given discrete linearization Φ and process noise $Q = L_Q L_Q^\mathsf{T}$, the covariance prediction is $P^+ = \Phi P \Phi^\mathsf{T} + Q$.

Lemma (QR square-root update). Let $A = \Phi L$ and form the stacked matrix

$$S = \begin{bmatrix} A^\mathsf{T} \\ L_Q^\mathsf{T} \end{bmatrix} \in \mathbb{R}^{2n\times n}. \tag{26}$$

Compute a thin QR factorization $S = QR$ with $Q^\mathsf{T}Q = I$ and R upper-triangular. Then $R^\mathsf{T}R = P^+$, and thus $L^+ := R^\mathsf{T}$ is a valid Cholesky factor of P^+.

Proof. By orthonormality of Q, $R^{\mathsf{T}}R = S^{\mathsf{T}}S$. But $S^{\mathsf{T}}S = AA^{\mathsf{T}} + L_Q L_Q^{\mathsf{T}} = \Phi L L^{\mathsf{T}}\Phi^{\mathsf{T}} + Q = P^+$. Hence $P^+ = R^{\mathsf{T}}R$ and $L^+ = R^{\mathsf{T}}$ satisfies $P^+ = L^+(L^+)^{\mathsf{T}}$. *QED*

This guarantees P^+ remains symmetric positive definite in floating-point arithmetic up to the conditioning of the QR.

7 FIXED-LAG RTS SMOOTHING (2 S WINDOW)

We implement a Rauch–Tung–Striebel smoother over the fixed-lag snapshot chain. Let (x_k, P_k) be filtered estimates and (Φ_k, Q_k) the stored propagation records between snapshots. Define $P_{k+1|k} = \Phi_k P_k \Phi_k^{\mathsf{T}} + Q_k$ and the smoother gain $A_k = P_k \Phi_k^{\mathsf{T}} P_{k+1|k}^{-1}$. The backward recursion is

$$x_k^s = x_k + A_k(x_{k+1}^s - x_{k+1|k}), \tag{27}$$

$$P_k^s = P_k + A_k(P_{k+1}^s - P_{k+1|k})A_k^{\mathsf{T}}. \tag{28}$$

Because the window is short (10 s max) and $n = 24$, this is computationally cheap and numerically stable with SPD enforcement.

8 COMPUTE BUDGET AND HOTSPOTS

8.1 Matrix dimensions and asymptotic cost

The estimator uses $n = 24$. Per IMU tick (400 Hz), the dominant cost is the SR time update via QR on a $(2n) \times n = 48 \times 24$ matrix: $O(n^3)$ with a small constant. GNSS PVT updates use $m = 6$ with $O(nm^2 + m^3)$. Tight-coupled raw GNSS uses m = rows ≤ 64 (typically much smaller).

8.2 Empirical timing (release build)

Measured on the current development machine (release build):

Kernel	iterations	time per iter
`propagate()` (includes SR update + history)	20,000	14,548 ns
`time_update_sqrt()` (isolated)	50,000	10,000 ns
`update_gnss()` (6D)	20,000	9,570 ns
`update_gps_raw()` (8 sats, 16 rows)	5,000	15,084 ns

These benchmarks establish a substantial margin against the 5 ms cycle budget. Direct timing equivalence to the target ARM A78AE processing core still requires hardware validation. The kernel execution times on the development machine remain in the microsecond range.

9 MONTE CARLO VERIFICATION AND RESULTS

9.1 Fault injection matrix (harness)

The harness simulates the full blackout mission environment. GNSS is removed over $[120, 160]$ s. Applicable sensors are subject to independent Bernoulli dropout with $p = 0.20$. Heavy-tailed outliers are

injected at a five percent rate on GNSS position, raw pseudorange, and radar altitude. The delayed measurement model uses a star-tracker timestamp of $t - 0.12$ s and a radar timestamp of $t - 0.07$ s, while raw GPS packets remain replay-safe if delivered late. A fault mode enabled through `BIAS_JUMP=1` imposes a step change in truth IMU bias at $t = 140$ s with random direction. The truth and filter paths also share the same deterministic RCS pulse train through the common nominal dynamics helper.

9.2 Metrics

We evaluate on the post-blackout acceptance window $[t_{\text{exit}}, t_{\text{exit}} + 2]$: RMS position, velocity, attitude error, and bias RMS (converted to µg and deg/s). We report per-run NEES/NIS means and their 95% coverage rates, plus phase-split NEES/NIS (pre/blackout/post/tail) to localize inconsistency to specific mission regimes.

9.3 Current quick regression (2026-06-13, 10 runs)

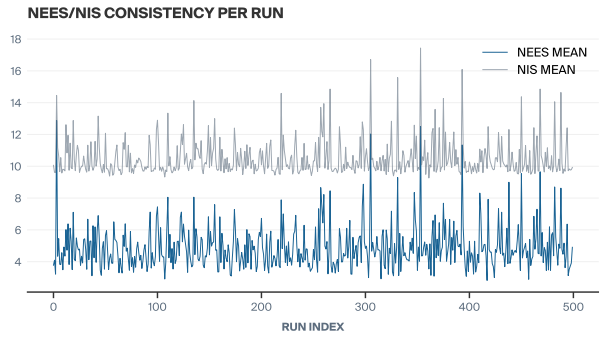
Mode	Pass rate	Max vel 3σ [m/s]	Mean NEES	Mean NIS
PVT baseline	100.0%	0.4157	4.80	5.89
Tight-coupled	100.0%	0.1044	4.82	11.69

9.4 Historical Monte Carlo summary (2026-03-11, 500 runs)

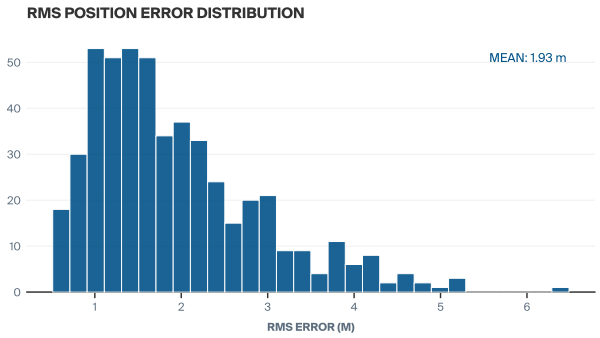
Mode	Pass rate	RMS pos [m]	RMS vel [m/s]	NEES mean / cov
PVT baseline	97.0%	1.50	0.097	134.49 / 0.945
Tight-coupled	100.0%	1.93	0.026	4.89 / 0.948

Interpretation. The historical 500-run record showed that tight coupling removed the rare catastrophic RV inconsistency events seen in the older PVT baseline. That baseline exhibited a heavy-tailed NEES distribution with 95th percentile ≈ 7.14 and maximum $\approx 3.18 \times 10^4$. The current repository state no longer reproduces those earlier acceptance failures in the quick regression above, but a fresh long campaign is still required before replacing the historical table with a new canonical Monte Carlo record.

9.5 Plots

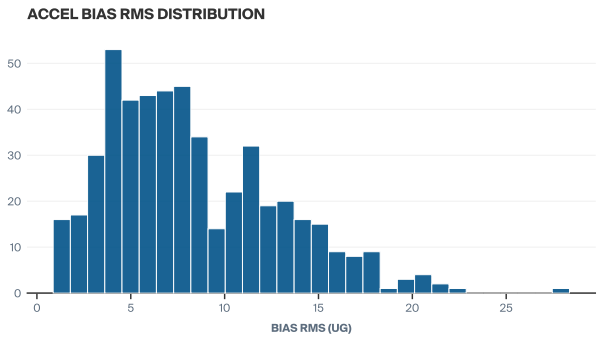


(a) Tight-coupled NEES/NIS per run.

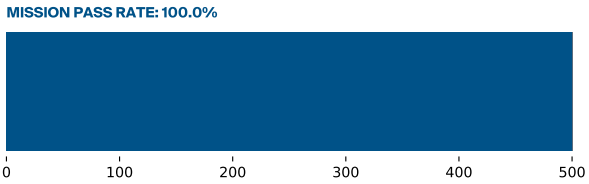


(b) Tight-coupled RMS position histogram.

Figure 1: Monte Carlo diagnostics for tight-coupled mode (500 runs).

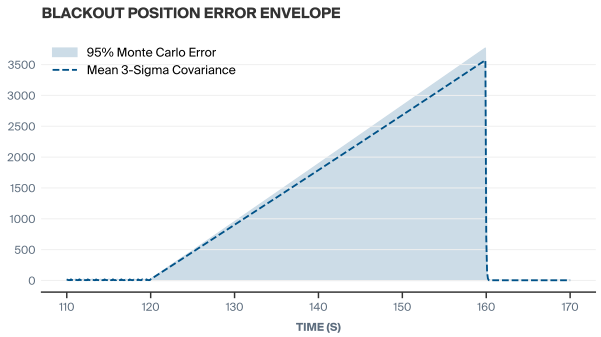


(a) Accel bias RMS distribution (tight).

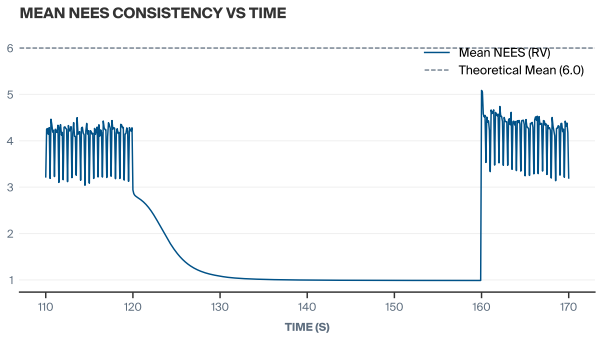


(b) Pass-rate indicator (tight).

Figure 2: Post-blackout window performance for tight-coupled mode.



(a) Position error envelope vs mean 3σ (500-run batch).



(b) Mean NEES(RV) vs time with 95% χ^2 band.

Figure 3: Blackout crossing diagnostics for tight-coupled mode (BLACKOUT_TRACE=1).

10 TRACEABILITY TO IMPLEMENTATION

The report maps directly onto the code base. Propagation and square-root time update are implemented in `src/estimator/filter/propagate.rs`. Process-noise construction and phase scaling are defined in `src/estimator/support/noise.rs` and `src/types.rs`. Delayed measurement rewind and replay are handled in `src/estimator/history/rewind.rs`. GNSS PVT update logic, including gating and covariance inflation, resides in `src/estimator/update/gnss.rs`. Tight-coupled raw GNSS processing, including robust pseudo-range handling, resides in `src/estimator/update/gps_raw.rs`. The fixed-lag RTS smoother is implemented in `src/estimator/filter/rts.rs`. Benchmark execution, plotting, and exported diagnostics are driven by `src/bin/harness.rs` and `src/plot.rs`.

11 CONCLUSION

The current repository implements a tight-coupled square-root error-state EKF with delay-aware rewind–update–replay, fixed-lag smoothing, phase-dependent covariance control, and pre-blackout bias calibration. The March 11, 2026 benchmark record remains a useful historical baseline, showing a 100.0% pass rate in the 500-run tight-coupled Monte Carlo campaign and a 97.0% pass rate in the PVT baseline under the prescribed blackout, dropout, and outlier conditions. A fresh full campaign is required to claim a current validated record under the aligned acceptance logic.

The remaining work is well bounded. The most useful next steps are tighter per-phase consistency control, especially during blackout where NEES remains conservative, and explicit bias step-fault campaigns

to complete the fault-injection matrix.